

Life Cycle Policies

1. Create a table comparing Amazon EBS and Amazon S3, listing at least 4 key differences (such as data type, access method, use case, and scalability) and give one real-world app example where each would be the better choice.

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Feature	Amazon EBS (Elastic Block Store)	Amazon S3 (Simple Storage Service)
Data Type	Block storage used as virtual hard disks for EC2 instances.	Object storage used to store files such as images, videos, documents, and backups.
Access Method	Can only be attached to EC2 instances within the same Availability Zone.	Accessible through the internet using the AWS Console, SDKs, CLI, or REST APIs.
Primary Use Case	Operating systems, databases, application files, and boot volumes.	Static website content, media files, backups, logs, and archives.
Scalability	Can be resized, but each volume is attached to an EC2 instance.	Virtually unlimited storage with automatic scalability.
Performance	High-performance, low-latency storage for applications requiring frequent read/write operations.	Optimized for storing and retrieving large numbers of objects rather than block-level access.
Real-World Example	Flipkart uses block storage for database servers that store customer orders and transactions.	Instagram stores profile pictures, photos, and videos in Amazon S3 because it is highly scalable and durable.

2. Launch a new EC2 instance in the AWS Free Tier and attach a new EBS volume of at least 2 GB to it using the AWS Management Console.

Hint: After launching the instance, use the 'Volumes' section in EC2 to create and attach the volume.

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1. Log in to the **AWS Management Console**.
2. Open the **EC2 Dashboard**.
3. Launch a new **Amazon Linux 2** or **Ubuntu** Free Tier instance.

4. Open **EC2** → **Elastic Block Store** → **Volumes**.
 5. Click **Create Volume**.
 6. Configure:
 - Size: **2 GB**
 - Type: **gp3 (General Purpose SSD)**
 - Availability Zone: Same as the EC2 instance.
 7. Click **Create Volume**.
 8. Select the volume and choose **Actions** → **Attach Volume**.
 9. Select your EC2 instance and attach the volume.
3. Log in to your EC2 instance using SSH and format the attached EBS volume, then mount it to a new directory named /mnt/mydata.

Hint: Use lsblk to find your volume, mkfs to format, and mount to attach it to the directory.

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- 1) Connect to the EC2 instance using SSH.
 - List available disks:
 - lsblk
 - 2) Format the new EBS volume:
 - sudo mkfs -t ext4 /dev/xvdf
 - 3) Create a mount point:
 - sudo mkdir /mnt/mydata
 - 4) Mount the volume:
 - sudo mount /dev/xvdf /mnt/mydata
 - 5) Verify:
 - df -h

➤ The mounted EBS volume should appear as **/mnt/mydata**.
4. Simulate a Zomato-style image upload: create a text file named zomato_menu.txt on your EC2 instance, copy it to your mounted EBS volume, and verify it is accessible from the new mount point.

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- 1) Create a text file:
 - echo "Zomato Restaurant Menu" > zomato_menu.txt
- 2) Copy it to the EBS volume:
 - sudo cp zomato_menu.txt /mnt/mydata/

- 3) Verify the file:
 - `ls /mnt/mydata`
 - 4) Display the contents:
 - `cat /mnt/mydata/zomato_menu.txt`
 - 5) Expected Output:
 - Zomato Restaurant Menu
5. Detach the EBS volume from your EC2 instance using the AWS Console, then reattach it to the same or a different instance. Describe what happens to the data you stored and why this behavior is useful for apps like Flipkart or Myntra order tracking.

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Open EC2 → Volumes.

1. Select the attached EBS volume.
2. Choose Actions → Detach Volume.
3. After detaching, attach it to the same or another EC2 instance in the same Availability Zone.
4. Mount the volume again using:

- `sudo mount /dev/xvdf /mnt/mydata`

❖ Observation

- The file `zomato_menu.txt` remains available after the EBS volume is detached and attached again. This is because Amazon EBS provides persistent block storage, meaning the data is stored independently of the EC2 instance.

❖ Real-World Benefit

- This feature is useful for applications like Flipkart or Myntra, where customer order data and transaction records must remain available even if an EC2 instance fails or is replaced. By attaching the existing EBS volume to a new instance, the application can quickly recover without losing important data.